

Problem 1.

$X = \{a, b, c\}$. The collection of subsets of X is

$$\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}.$$

There are 8 subsets.

Prove by induction that the number of subsets of an n -element set is 2^n .

Base case: $n=1, 2, 3$.

Inductive step. Let X be an $(n+1)$ -element set. WLOG, $X = \{1, 2, \dots, n+1\}$.

Let A' be the collection of subsets of $X' = \{1, 2, \dots, n\}$. Consider

$\mathcal{A} = \{A, A \cup \{n+1\} : A \in A'\}$, which is the collection of all subsets of

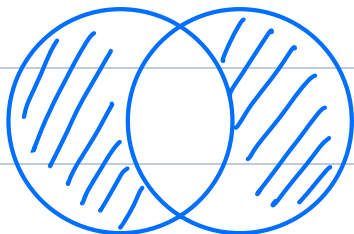
X . (Why? If $A \subseteq X$ but $n+1 \notin A$, then $A \subseteq X'$. Hence $A \in A' \subseteq \mathcal{A}$.

If $n+1 \in A$, then $A \setminus \{n+1\} \subseteq X'$; $A \setminus \{n+1\} \in A'$.

Still we have $A \in \mathcal{A}$).

$$|\mathcal{A}| = 2|A'| = 2^{n+1}.$$

Problem 2



$$A \Delta B$$

A B

$$(A \Delta B) \Delta C = \text{Venn diagram with three circles A, B, C. The regions of A, B, and C that do not overlap with each other are shaded with diagonal lines. The intersection of A and B is shaded, the intersection of B and C is shaded, and the intersection of A and C is shaded. The intersection of all three is shaded. The regions of A, B, and C that do not overlap with each other are shaded.} = A \Delta (B \Delta C)$$

Problem 3

(i) $g \circ f$ is an injection:

If $x_1 \neq x_2 \in X$, then $f(x_1) \neq f(x_2) \in Y$, and $g(f(x_1)) \neq g(f(x_2))$.

$\uparrow$ f is injection.

$\uparrow$ g is surjection.

$g \circ f$ is a surjection:

For any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$.

$\uparrow$ g is a surjection.

There exists $x \in X$ such that $f(x) = y$. $\leftarrow f$ is a surjection.

Hence, $g \circ f(x) = z$.

(ii) f^{-1} is a surjection: for any $x \in X$, $f(x) \in Y$, and $f^{-1}(f(x)) = x$.

f^{-1} is an injection: If $y_1 \neq y_2$, then definitely $f^{-1}(y_1) \neq f^{-1}(y_2)$,

as otherwise $y_1 = f(f^{-1}(y_1)) = f(f^{-1}(y_2)) = y_2$.

$$f^{-1} \circ f = f \circ f^{-1} = \text{id}_X \Rightarrow (f^{-1})^{-1} = f.$$

(iii) $\text{id}_X = f \circ f^{-1}$ and apply (ii) & (i).

Problem 4. (a) Let $g(x) = \frac{x}{1-|x|}$ Then

$$f \circ g(x) = \frac{\frac{x}{1-|x|}}{1 + \left| \frac{x}{1-|x|} \right|} = \frac{x}{1 - |x| + |x|} = x.$$

$$g \circ f(x) = \frac{\frac{x}{1+|x|}}{1 - \left| \frac{x}{1+|x|} \right|} = \frac{x}{1 + |x| - |x|} = x.$$

Hence, $f \circ g = \text{id}_{(-1,1)}$ and $g \circ f = \text{id}_{\mathbb{R}}$.

$\Rightarrow f$ is bijection since f has left & right inverse.

(b)
$$s(x) = \begin{cases} \sqrt{x} & x \geq 0 \\ -\sqrt{-x} & x < 0 \end{cases}.$$

Easy to check that $s \circ h = \text{id}_{\mathbb{R}}$. Hence h is injective.

Similarly, $h \circ s = \text{id}_{\mathbb{R}}$. h is surjective.